



HOCHSCHULE FÜR TECHNIK UND WIRTSCHAFT

M7 : MANUFACTURING TECHNOLOGY AND BUSINESS TECHNOLOGY

Project Management Report- MM4 - CellTech Solutions(Pouch Cell)

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July 12, 2023

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Chapter 1

Introduction

This report summarizes the development in making of Pouch Cell in a factory incorporating the use of Technomatix Plant Simulator, under the guidance of Prof. Christoph Sprung. We have completed the pre-production phase, got an idea of how Pouch cell is manufactured and reached our first milestone. Now that we understand the basics and have created cell of one size with one cell chemistry in simulation with set parameters, we have now added two more cell sizes to achieve our second milestone. For our third milestone we have described the tasks and assigned them. We have successfully completed the task allotted as per Milestone-3 by achieving OEE of 70% in simulation, created data sheet which have details of health and safety, Quality and standards needed to be set for a factory to be safe and hazardous free. Moving ahead with Milestone-4, we have given the details about how we have planned to achieve it in this report.

Chapter 2

Work Breakdown Structure

Work Breakdown Structure (WBS)

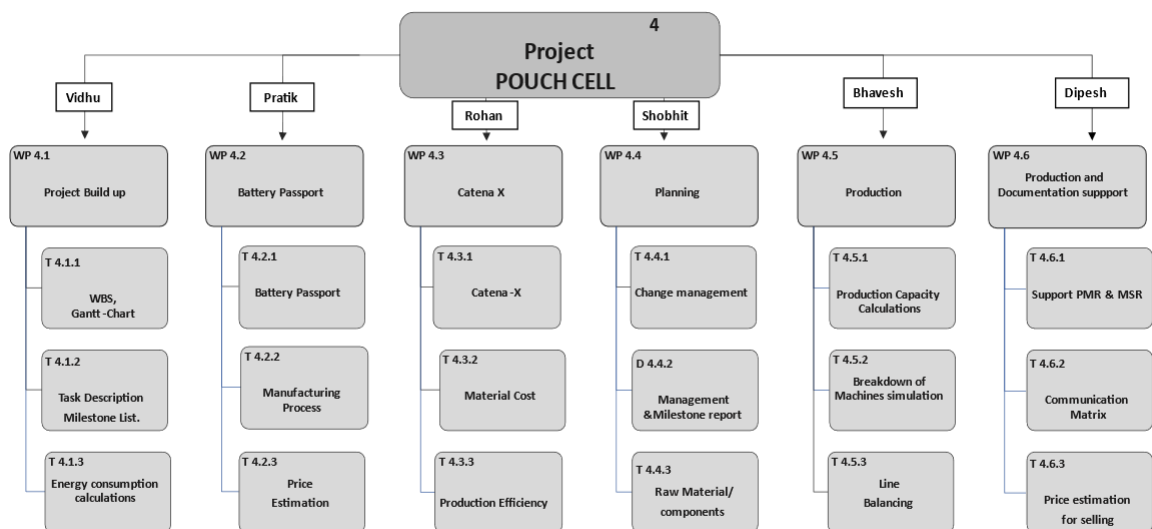


Figure 2.1: Work Breakdown Structure

As per the comfort and previous experience work packages has been allocated to all the team members as given in the Figure:2.1 Work Breakdown Structure.

Work Breakdown Structure (WBS): Create a detailed breakdown of tasks and subtasks required

for battery manufacturing optimization.

2.1 Vidhu's Task

T 4.1.1:

Create a detailed breakdown of tasks and subtasks required for battery manufacturing optimization. Develop a Gantt chart to visualize the project schedule, including task dependencies and timelines.

T 4.1.2:

Provide detailed descriptions for each task in the project plan, outlining the specific objectives and deliverables.

T 4.1.3:

Analyse the energy consumption of the project.

Calculate the energy requirements for different stages or processes. Identify potential areas for energy optimization and efficiency improvements.

2.2 Pratik's Tasks:

T 4.2.1: Develop a standardized battery passport system to track the manufacturing history and performance of each battery.

T 4.2.2: Optimize the battery manufacturing process by identifying bottlenecks, streamlining procedures, and implementing efficiency improvements.

T 4.2.3: Conduct a comprehensive analysis to determine the project's manufacturing costs. Consider factors such as raw materials, labour, overheads, and equipment expenses. Determine the selling price based on cost estimation, market trends, and profit margin goals.

2.3 Rohan's Tasks:

T 4.3.1: Implement the Catena X system for battery manufacturing, which includes data acquisition, analysis, and process control.

T 4.3.2: Calculate the cost of materials required for the project. Consider factors such as quantity, quality, and current market prices. Optimize material usage to minimize costs while maintaining product quality.

T 4.3.3: Analyse the potential variations in production efficiency due to different parameters. Identify the critical parameters affecting production efficiency and their potential variations. De-

velop strategies to optimize production efficiency and mitigate the effects of parameter.

2.4 Shobhit's Tasks:

T 4.4.1: Identify and manage potential changes in project scope, schedule, and resources, ensuring minimal disruption to the manufacturing process.

D 4.4.2: Regularly update and present management and progress reports, highlighting key achievements, challenges, and recommendations.

T 4.4.3: Coordinate and oversee the production of raw materials or components required for the project. Ensure timely delivery and quality control of raw materials or components. Maintain inventory levels and address any production-related issues.

2.5 Bhavesh's Tasks:

T 4.5.1: Calculate the maximum production capacity of the project's manufacturing facility. Consider factors such as machine capabilities, labour availability, and production cycle times. Determine the production capacity utilization rate.

T 4.5.2: Conduct simulations to analyse the breakdown patterns of machines in the production process. Identify potential bottlenecks and develop contingency plans to minimize downtime. Optimize maintenance schedules and implement predictive maintenance techniques.

T 4.5.3: Analyse production demand variations and their impact online balancing. Optimize the allocation of tasks across production lines to maximize efficiency and minimize idle time. Develop strategies for flexible line balancing to accommodate changing production demands.

2.6 Dipesh's Tasks:

T 4.6.1: Help and contribute to the preparation of both the project report and management report, ensuring accuracy and completeness.

T 4.6.2: Develop a communication matrix to facilitate effective communication and coordination among team members.

T 4.6.3: Determine the selling price of the project's product based on market analysis and cost estimation. Consider factors such as competition, customer demand, and value proposition. Evaluate pricing strategies and establish a pricing model to maximize profitability.

Detailed description of what tasks are being taken by which member is given in, "DESCRIPTION OF TASKS."

Chapter 3

Gantt Chart

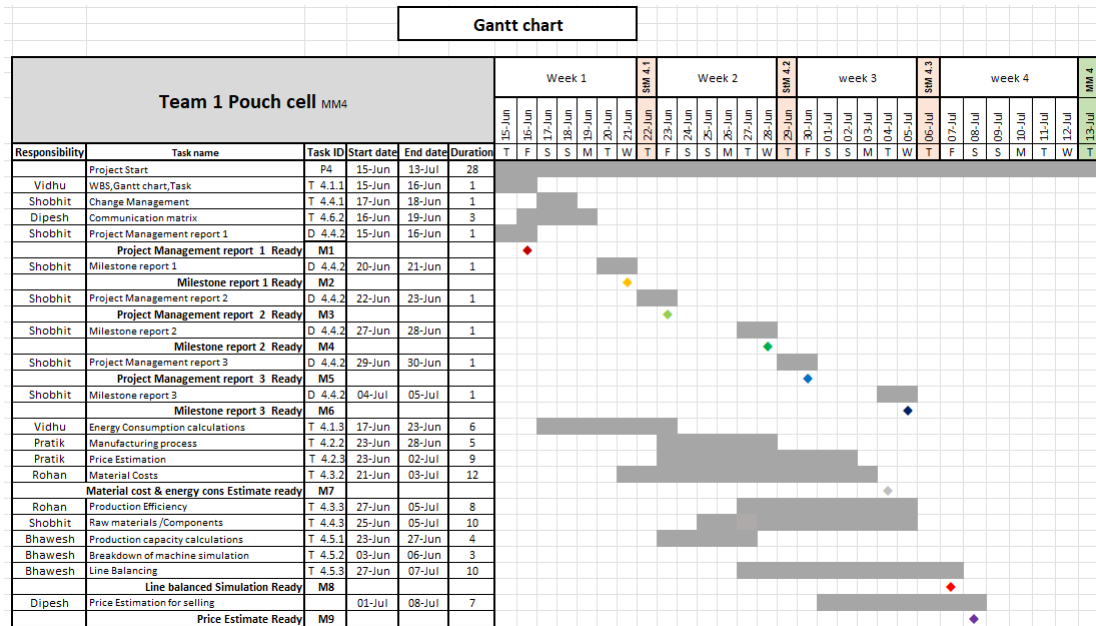


Figure 3.1: Gantt Chart

Gantt Chart describes given time to a particular task. As presented in the Figure 3.1: Gantt Chart, various milestone along with different categories are divided throughout the timeline of the semester to understand the time needed for all the tasks.

According to the Gantt Chart, the team has now worked on the Pre-Production phase, which covered Software knowledge, Understanding the project, and Documentation. Bhavesh obtained software knowledge because of this. Pratik finished the battery information (passport). Work packages have been assigned, and everyone understands their roles and responsibilities.

We have finalized the material and battery composition. Along with chemical composition we have also confirmed the size of the battery. Given the details we have, we have worked and completed our first simulation to have an idea of how we can analyse and use that data to optimize the simulation for effective results.

By the end of Milestone 2 we have multiplied our production per day by 4 times with one cell chemistry and three different sizes.

Above we can see we have finished working on our tasks to achieve Milestone-3.

Gantt chart is updated for MM4

Chapter 4

Milestone Trend Analysis

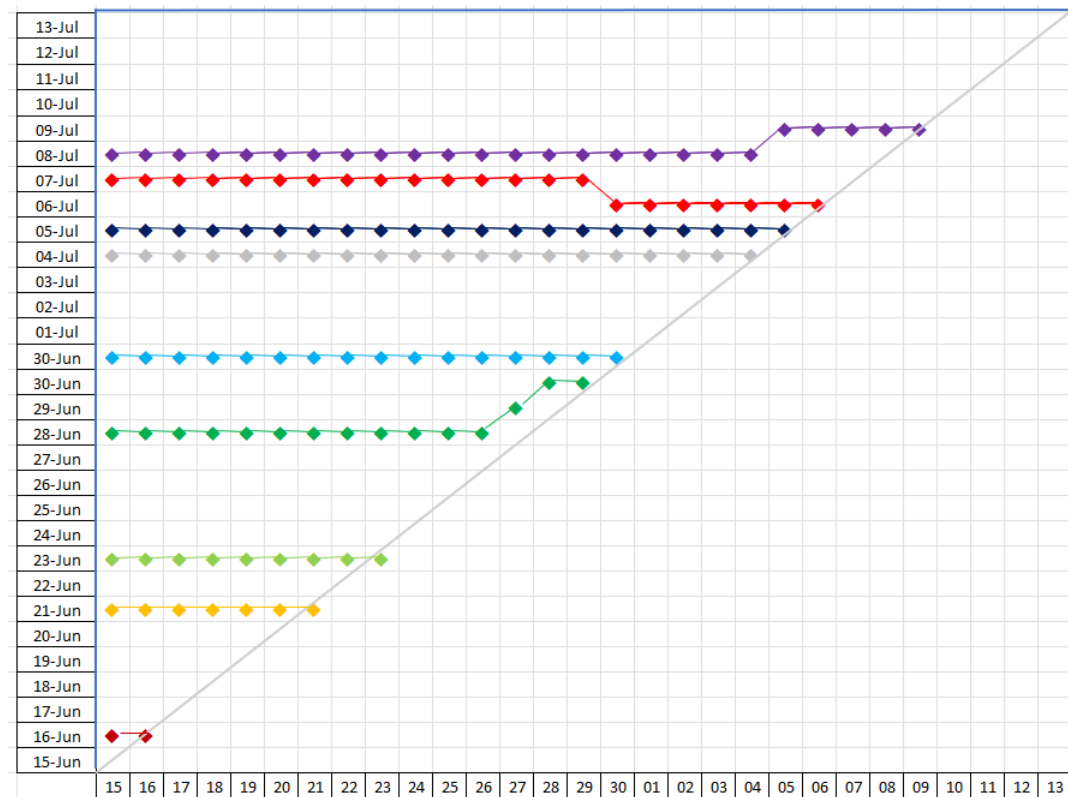


Figure 4.1: Milestone Trend Analysis

Milestones list & Analysis			Date	Status
M1	Project Management Report 1	◆	16-Jun	Completed
M2	Milestone Report 1	◆	21-Jun	Completed
M3	Project Management Report 2	◆	23-Jun	Completed
M4	Milestone Report 2	◆	28-Jun	Completed
M5	Project Management Report 3	◆	30-Jun	Completed
M6	Milestone Report 3	◆	05-Jul	Completed
M7	Material cost & energy cons	◆	04-Jul	Completed
M8	Line balanced Simulation	◆	07-Jul	Completed
M9	Price Estimate	◆	08-Jul	Completed

Figure 4.2: Milestone Trend analysis Abbreviations with dates

4.1 Milestone 1,3,5

Project Management Report

Completion and submission of a project management report. The report will outline the overall project plan, including the objectives, scope, timeline, resource allocation, and key milestones. It will also include details on the project team, roles and responsibilities, and communication strategies.

4.2 Milestone 2,3,6:

Milestone Report

Document the progress and achievements made up to this point in the project. It will provide an overview of the tasks completed, milestones achieved, and any challenges or risks encountered. The report will also include a review of the project and ensuring that the project is on track.

4.3 Milestone 7:

Material Cost and Energy Consumption Estimate

Estimating the material costs and energy consumption associated with our manufacturing process like the various materials used in the production process, their quantities, and associated costs. Additionally, analyse the energy requirements and consumption throughout the production stages, providing insights into potential areas for optimization and cost reduction.

4.4 Milestone 8:

Line Balance Simulation

Analysing and optimizing the production line's efficiency and productivity, evaluating the flow of materials, equipment utilization, and cycle times. It will identify any areas of inefficiency and propose solutions to balance the production line, optimize resource utilization, and enhance overall productivity.

4.5 Milestone 9:

Price Estimate

Estimating the price for the pouch cell by considering factors such as material costs, labour costs, overhead expenses, and desired profit margins. It will also account for market conditions, competition, and customer expectations. The price estimate will help in setting competitive and sustainable pricing strategies for the pouch cell products.

Chapter 5

Communication Matrix

COMMUNICATION MATRIX												
Date	15-Jun	18-Jun	21-Jun	22-Jun	25-Jun	28-Jun	29-Jun	2-Jul	5-July	6-July	9-July	12-July
Meeting Type	Face-to-Face	Virtual (Google Meet)	Face-to-Face	Face-to-Face	Virtual (Google Meet)	Face-to-Face	Face-to-Face	Virtual (Google Meet)	Face-to-Face	Face-to-Face	Virtual (Google Meet)	Face-to-Face
Medium	Email / Whatsapp	Email / Whatsapp	Email / Whatsapp	Email / Whatsapp	Email / Whatsapp	Email / Whatsapp	Email / Whatsapp	Email / Whatsapp	Email / Whatsapp	Email / Whatsapp	Email / Whatsapp	Email / Whatsapp
Discussions	• Planing of Next Milestone • Task allocation Planning	• Update of Task Assigned • Planning and assigning next Tasks	• Report Review and Discussions	• Planinng of Next Milestone • Task allocation Planning	• Update of Task Assigned • Planning and assigning next Tasks	• Report Review and Discussions	• Planing of Next Milestone • Task allocation Planning	• Update of Task Assigned • Planning and assigning next Tasks	• Report Review and Discussions	• Planinng of Next Milestone • Task allocation Planning	• Update of Task Assigned • Planning and assigning next Tasks	• Report Review and Discussions
Output	• Tasks Assigned Accordingl y	• Preparing Progress Report • Sending Progress Report	• Report Reviewed, Approved and Sent	• Tasks Assigned Accordingl y	• Preparing Progress Report • Sending Progress Report	• Report Reviewed, Approved and Sent	• Tasks Assigned Accordingl y	• Preparing Progress Report • Sending Progress Report	• Report Reviewed, Approved and Sent	• Tasks Assigned Accordingl y	• Preparing Progress Report • Sending Progress Report	• Report Reviewed, Approved and Sent

Figure 5.1: Communication Matrix MM4 (15th June 2023- 12th July 2023)

TASK ALLOCATION						
Task Allocated for	Bhavesht Tawade	Pratik Wankhede	Rohan Sahu	Shobhit Pise	Vidhu Sreekumar	Dipesh Gawali
Task Allocated By	Pratik	Shobhit	Shobhit	Pratik	Pratik and Shobhit	Pratik and Shobhit
Task allocated on Date 15-June	- Production Capacity Calculations and Increase in Capacity. - Breakdown of Machines Simulation.	- Price estimation for selling the Product to the Market. - Deal with all things related to manufacturing process.	- Material Cost Calculations. - Production Efficiency Description.	- Raw Material/Component's Production. - Task allocation for every member.	- Energy Consumption Calculations. - Gantt Chart, WBS Updates.	- Production Parameter Variation. - Price estimation for selling the Product to the Market.
Task allocated on Date 22-June	- Production Capacity Calculations and Increase in Capacity. - Analyse production demand variations and their impact online balancing.	- Develop battery passport system to track manufacturing history and performance. - Optimize the battery manufacturing process.	- Implement Catena-X. - Calculate the cost of materials required for the project.	- Identify and manage changes in project scope, schedule, and resources. - Regularly update and present management and progress reports.	- Analyse the energy consumption of the project - Gantt Chart, WBS Updates.	- Determine the selling price of the project's product. - Contribute to the preparation of both the project report and management report.
Task allocated on Date 29-June	- Show Cycle Time in Calculations. - Explain impact of Breakdown. - Failure detection and mitigate it.	- Update Calculations. - Make Sankey Diagram for the process. - Help Bhavesht with Tasks.	- Scrap and wastage Calculations. - Slurry Calculations in cubic millimeter.	- CatenaX updates. - Changes in Report format e.g. Table citations, etc.	- Gantt Chart Updates. - WBS Updates.	- Explain Processes. - Viscosity Comparison. - Process Figures and Diagrams update.
Task allocated on Date 6-July	- Calculates the Kg/Min for each Machine and Percentage of Wastage. - Space required for Ageing.	- Sankey Diagram. - Explanation of Sankey Diagram.	- Implement Catena-X. - Update data where required.	- CatenaX updates. - Changes in Report format e.g. Table citations, etc.	- Slot die Calculations. - Power Consumptions table changes and updates.	- Hand drawn sketches of Rolling knives and welding. - Communication Matrix

Figure 5.2: Task allotment MM4 (15th June 2023- 12th July 2023)

All the meetings held and there details are give above.

For Milestone 4 total four feedback meetings were held. The meetings were held every Thursdays. For planning and discussion purpose 3 meeting each week were planned. Every Thursday tasks were discussed and assigned accordingly.

Following Sundays, the updates on the given tasks were given and any further plans were discussed. Every Wednesdays the information was compiled and added in the report to receive feedback.

For week one 15th June to 21st June tasks were planned and completed on time no delay in the set time was observed. Major planning and allocations of the tasks was done as new requirement were needed to be done.

For week two 22nd June to 28th June tasks were planned and assigned some additional tasks were added to update any failures or pending works. Due to task being unclear a little delay in the completion of the tasks was observed. Following two weeks no delay was seen each and ever task assigned was completed on time.

Chapter 6

Description of Tasks

6.1 Vidhu

Tasks allotted:

6.1.1 T.4.1.1

Objective: The objective of battery manufacturing optimization is to streamline processes, maximize efficiency, and minimize waste in order to improve overall productivity and product quality.

Results: The expected results of battery manufacturing optimization include reduced cycle times, minimized idle time, increased production capacity, improved product quality, and enhanced resource utilization.

Controlling: To ensure successful implementation and control, measures such as data collection and analysis, performance tracking of key metrics, regular reviews and audits, and timely corrective actions are implemented.

Resources: Moodle “Presentation Milestone Meeting 0 (MM0)”

Duration: 16/06/2023- 13/07/2023

Connection: The optimization process is connected to various areas such as process mapping, task allocation, equipment optimization, quality control, and continuous improvement initiatives to ensure alignment and holistic improvement throughout the battery manufacturing system.

6.1.2 T.4.1.2

Objective: The objective of the project plan is to optimize battery manufacturing processes by implementing specific tasks that address process inefficiencies and maximize productivity.

Results: The expected results of the project plan include streamlined processes, increased production efficiency, reduced cycle times, improved product quality, and enhanced resource utilization.

Controlling: To ensure successful implementation and control, measures such as regular progress tracking, milestone reviews, performance evaluations, and timely adjustments are implemented to monitor and steer the project towards its objectives.

Resources: NA

Duration: 16/06/2023- 13/07/2023

Connection: The project plan is connected to various areas such as process analysis, task optimization, quality control, resource management, and continuous improvement initiatives to ensure alignment and integration of efforts throughout the battery manufacturing optimization project.

6.1.3 T.4.1.3

Objective: The objective of the project plan is to optimize battery manufacturing processes by implementing specific tasks that address process inefficiencies and maximize productivity.

Results: The expected results of the project plan include streamlined processes, increased production efficiency, reduced cycle times, improved product quality, and enhanced resource utilization.

Controlling: To ensure successful implementation and control, measures such as regular progress tracking, milestone reviews, performance evaluations, and timely adjustments are implemented to monitor and steer the project towards its objectives.

Resources: Life cycle assessment of the energy consumption and GHG emissions of state-of-the-art automotive battery cell production by Florian Degen, Marius Schütte

Duration: 16/06/2023- 25/06/2023

Connection: The project plan is connected to various areas such as process analysis, task optimization, quality control, resource management, and continuous improvement initiatives to ensure alignment and integration of efforts throughout the battery manufacturing optimization project.

6.2 Pratik

Tasks allotted:

6.2.1 T.4.2.1

Objective: The objective of developing a standardized battery passport system is to establish a tracking mechanism that records the manufacturing history and performance data of each battery throughout its lifecycle.

Results: The expected results of implementing a battery passport system include improved traceability, enhanced quality control, better warranty management, and the ability to track and analyze battery performance for optimization.

Controlling: To ensure effective control, measures such as implementing a secure database or digital platform, establishing standardized data collection procedures, implementing quality checks and audits, and defining access controls are put in place to ensure the accuracy and integrity of the battery passport system.

Resources: N/A

Duration: 16/06/2023- 13/07/2023

Connection: The battery passport system is connected to various areas such as manufacturing, quality assurance, customer service, and research and development to facilitate data exchange, inform decision-making processes, support product recalls or warranty claims, and enable continuous improvement initiatives based on the gathered information.

6.2.2 T.4.2.2

Objective: The objective is to optimize the battery manufacturing process by identifying bottlenecks, streamlining procedures, and implementing efficiency improvements to enhance productivity and reduce costs.

Results: The expected results of the optimization efforts include increased production efficiency, reduced cycle times, improved product quality, cost savings, and enhanced resource utilization.

Controlling: To ensure effective control, measures such as process mapping, data analysis, performance tracking, regular reviews, and continuous improvement initiatives are implemented to monitor and steer the optimization process.

Resources: N/A

Duration: 16/06/2023- 13/07/2023

Connection: The optimization process is connected to various areas such as process analysis, task allocation, equipment optimization, quality control, and continuous improvement initiatives to ensure alignment and integration of efforts throughout the battery manufacturing process.

6.2.3 T.4.2.3

Objective: The objective of conducting a comprehensive cost analysis is to determine the manufacturing costs of the project by considering various factors such as raw materials, labor, overheads, and equipment expenses, and to establish a selling price based on cost estimation, market trends, and profit margin goals.

Results: The expected results of the cost analysis include a detailed breakdown of manufacturing costs, identification of cost drivers, determination of a competitive selling price that considers market trends and profit margin goals, and improved financial planning and decision-making.

Controlling: To ensure effective control, measures such as accurate data collection, cost tracking systems, regular cost reviews, monitoring of market trends, and profitability analysis are implemented to assess and control manufacturing costs and pricing decisions.

Resources: N/A

Duration: 16/06/2023- 13/07/2023

Connection: The cost analysis is connected to various areas such as procurement, production planning, financial management, and marketing to ensure alignment of cost estimation, pricing strategy, and overall business objectives.

6.3 Rohan

Tasks allotted:

6.3.1 T.4.3.1

Objective: Implementation of Catena- X.

Results: To start understanding opportunity of how Catena- X can be used to our profit.

Controlling: Planning on creation of roadmap and basic user interface.

Resources: Catena-X's official website.

Duration: 16/06/2023- 13/07/2023

Connection: To have knowledge about automotive suppliers and OEMs, enhance business opportunities and global reach. Catena-X will be needed.

6.3.2 T.4.3.2

Objective: The objective is to calculate the cost of materials required for the project by considering factors such as quantity, quality, and current market prices, and to optimize material usage to minimize costs while maintaining product quality.

Results: The expected results of the cost calculation and material optimization efforts include a detailed breakdown of material costs, identification of cost-saving opportunities through efficient material usage, improved cost control, and maintained product quality standards.

Controlling: To ensure effective control, measures such as accurate material quantity estimation, monitoring of market prices, vendor management, quality control checks, and regular reviews of material usage are implemented to assess and control material costs and optimize usage.

Resources: N/A

Duration: 16/06/2023- 13/07/2023

Connection: The cost calculation and material optimization process are connected to various areas such as procurement, production planning, quality assurance, and financial management to ensure alignment of material costs, production efficiency, product quality, and overall cost-saving objectives.

6.3.3 T.4.3.3

Objective: The objective is to analyze potential variations in production efficiency due to different parameters, identify critical parameters affecting production efficiency and their potential variations, and develop strategies to optimize production efficiency and mitigate the effects of these parameters.

Results: The expected results of the analysis and optimization efforts include a comprehensive understanding of parameter variations and their impact on production efficiency, identification of critical parameters, implementation of strategies to optimize efficiency, reduced variability in production processes, and improved overall productivity.

Controlling: To ensure effective control, measures such as data collection and analysis, monitoring of parameter values, implementation of process control mechanisms, performance tracking of key indicators, and regular reviews and adjustments are implemented to control and optimize production efficiency.

Resources: N/A

Duration: 16/06/2023- 13/07/2023

Connection: The analysis and optimization of production efficiency are connected to various areas such as process analysis, equipment optimization, quality control, workforce management, and continuous improvement initiatives to ensure alignment and integration of efforts in maximizing efficiency and productivity.

6.4 Shobhit

Tasks allotted:

6.4.1 T.4.4.1

Objective: The objective is to identify and manage potential changes in project scope, schedule, and resources in order to ensure minimal disruption to the manufacturing process.

Results: The expected results of identifying and managing potential changes include timely recognition of scope, schedule, and resource adjustments, effective change management processes, minimized disruptions to the manufacturing process, and successful project completion within established parameters.

Controlling: To ensure effective control, measures such as regular monitoring of project progress, proactive identification of potential changes, implementing change request processes, resource allocation adjustments, and risk management strategies are implemented to control and mitigate the impact of changes on the manufacturing process.

Resources: NA

Duration: 26/05/2023- 15/06/2023

Connection: The identification and management of potential changes are connected to various areas such as project management, production planning, resource allocation, and communication channels to ensure alignment and collaboration among stakeholders, enabling seamless integration of changes while minimizing disruptions to the manufacturing process.

6.4.2 D.4.4.2

Objective: The objective is to regularly update and present management and progress reports, highlighting key achievements, challenges, and recommendations to ensure effective communication, transparency, and informed decision-making.

Results: The expected results of updating and presenting management and progress reports include a clear overview of project status, identification of key accomplishments, highlighting challenges and issues, providing informed recommendations for decision-making, and fostering effective communication between project stakeholders.

Controlling: To ensure effective control, measures such as establishing reporting schedules, defining report templates and formats, collecting accurate and relevant data, conducting thorough analysis, and incorporating feedback from management and stakeholders are implemented to maintain the quality and consistency of management and progress reports.

Resources:NA

Duration: 16/06/2023- 13/07/2023

Connection:The regular updating and presentation of management and progress reports are connected to various areas such as project management, stakeholder engagement, decision-making processes, and performance evaluation to facilitate informed decision-making, enable timely interventions, and maintain alignment among project stakeholders.

6.4.3 T.4.4.3

Objective: The objective is to coordinate and oversee the production of raw materials or components required for the project, ensuring timely delivery, quality control, maintaining inventory levels, and addressing any production-related issues.

Results: The expected results of coordinating and overseeing production include efficient production of raw materials or components, timely delivery to meet project requirements, adherence to quality control standards, optimized inventory management, and effective resolution of production-related issues.

Controlling: To ensure effective control, measures such as production planning and scheduling, quality control procedures, supplier management, inventory tracking, performance monitoring, and timely issue resolution are implemented to maintain control over the production process and ensure the availability of materials or components.

Resources:NA

Duration: 16/06/2023- 13/07/2023

Connection: The coordination and oversight of production are connected to various areas such as procurement, production planning, quality assurance, inventory management, and project management to maintain synchronization and alignment of activities, ensuring smooth production operations and meeting project timelines.

6.5 Bhavesh

6.5.1 T 4.5.1

Objective: To calculate the maximum production capacity of the project's manufacturing facility.

Results: Determine the production capacity utilization rate.

Controlling: Machine capabilities, labour availability, and production cycle time

Resources:NA

Duration: 16/06/2023-13/07/2023

Connection:

6.5.2 T 4.5.2

Objective: Conduct simulations to analyse the breakdown patterns of machines in the production process.

Results:Improve maintenance plans and put predictive maintenance methods into practice.

Controlling: Simulation limitations.

Resources: NA

Duration: 16/06/2023- 13/07/2023

Connection: If simulation not optimized, optimum output will not be attend.

6.5.3 T 4.5.3

Objective: The objective of analyzing the energy consumption of the project is to identify areas for energy optimization and efficiency improvements throughout different stages or processes of battery manufacturing.

Results: The expected results of the energy consumption analysis include a comprehensive understanding of energy requirements for each stage or process, identification of potential areas for energy optimization, and recommendations for efficiency improvements to reduce energy consumption.

Controlling: To ensure effective control, measures such as data collection on energy consumption, analysis of energy usage patterns, implementation of energy monitoring systems, and tracking of key energy performance indicators are implemented to monitor and control energy usage.

Resources: NA

Duration: 16/06/2023- 13/07/2023

Connection: The analysis of energy consumption is connected to various areas such as process mapping, equipment optimization, task allocation, and sustainability initiatives to ensure alignment and integration of energy optimization efforts throughout the battery manufacturing project.

6.6 Dipesh

Tasks allotted:

6.6.1 T.4.6.1

Objective: The objective is to prepare both the project report and management report with a focus on accuracy and completeness.

Results: The results of the preparation process will include comprehensive and reliable project and management reports.

Controlling: The preparation process will be monitored and controlled to ensure adherence to quality standards and guidelines.

Resources:

Duration: 16/06/2023- 13/07/2023

Connection: The project report and management report will be interconnected, providing a cohesive overview of the project's progress and outcomes.

6.6.2 T.4.6.2

Objective: The objective is to develop a communication matrix to enhance communication and coordination among team members.

Results: The results of implementing the communication matrix will include improved information flow, increased collaboration, and enhanced team productivity.

Controlling: The communication matrix will be monitored and controlled to ensure its effectiveness in facilitating effective communication and coordination.

Resources: N/A

Duration: 16/06/2023- 13/07/2023

Connection: The communication matrix will serve as a central tool for team members to stay connected, share information, and coordinate their efforts efficiently.

6.6.3 T.4.6.3

Objective: The objective is to determine the selling price of the project's product by conducting a thorough market analysis and cost estimation, considering competition, customer demand, and value proposition, and to establish a pricing model that maximizes profitability.

Results: The results will include a well-informed and competitive selling price for the product, taking into account market dynamics and cost factors, leading to improved profitability.

Controlling: The pricing strategy and model will be closely monitored and controlled to ensure its effectiveness in maximizing profitability, with regular assessments and adjustments as necessary.

Resources: N/A

Duration: 16/06/2023- 13/07/2023

Connection: The pricing decisions will be connected to the market analysis, cost estimation, and value proposition, aligning the selling price with the target market's needs and preferences while optimizing profitability.